

Applying RNA velocity to infer cell fate transitions between naïve human ESCs and 8C-like cells

So what are 8C-like cells?

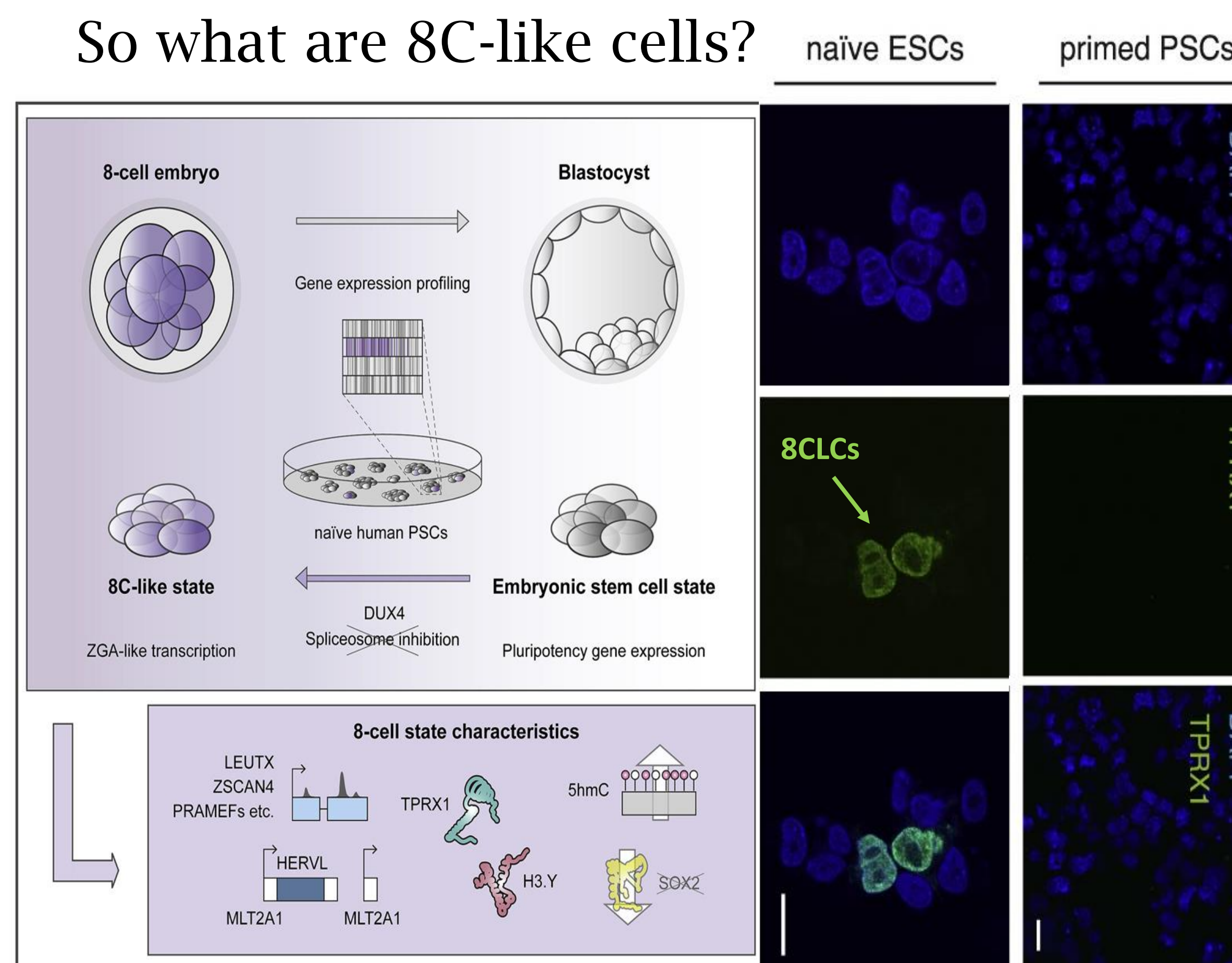


Fig 1: Description of 8CLCs. Taubenschmid-Stowers J, et al. 8C-like cells capture the human zygotic genome activation program in vitro. Cell Stem Cell. 2022;29(3):449-459.e6.

- Human Zygotic Genome Activation (**ZGA**), the first major **wave of transcription** after fertilization, occurs at the **8C stage**, leading to **loss of totipotency thereafter**
- 8C-like cells (8CLC) are an *in vitro* subpopulation of naïve human embryonic stem cells that resemble 8C embryos
- 8C embryos are inaccessible due to IVF cells being harvested at a later stage, making 8CLCs crucial to studying ZGA

What is RNA Velocity?

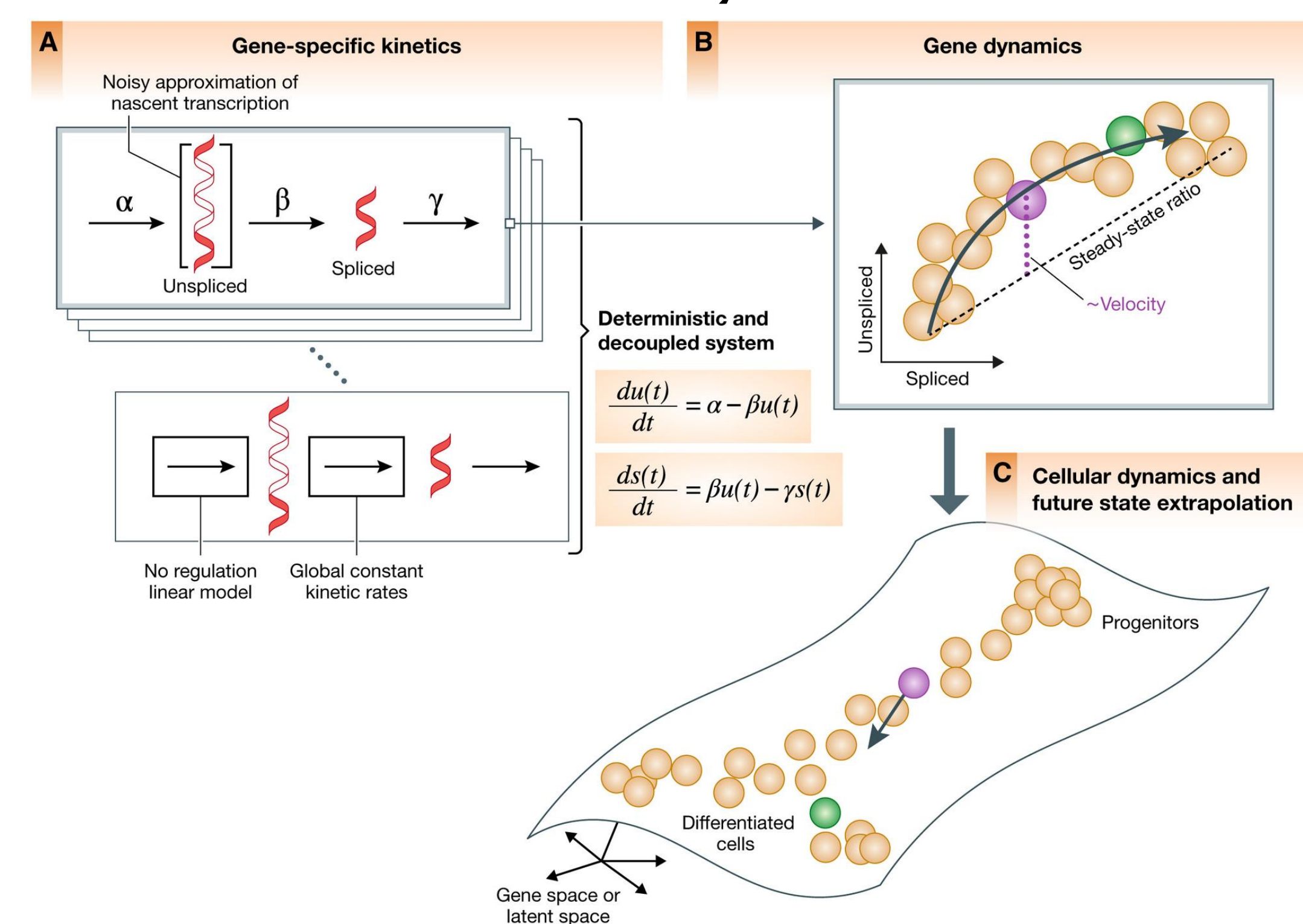


Fig 2: Description of RNA velocity. Bergen, V., et al. Generalizing RNA velocity to transient cell states through dynamical modeling. Nat Biotechnol 38, 1408–1414 (2020).

- Uses **unspliced vs spliced** scRNA-seq read ratios for each gene
- **scVelo predicts a “velocity”** for the transcriptome relative to the “positions”/steady states of each cell

Example ZGA driver genes

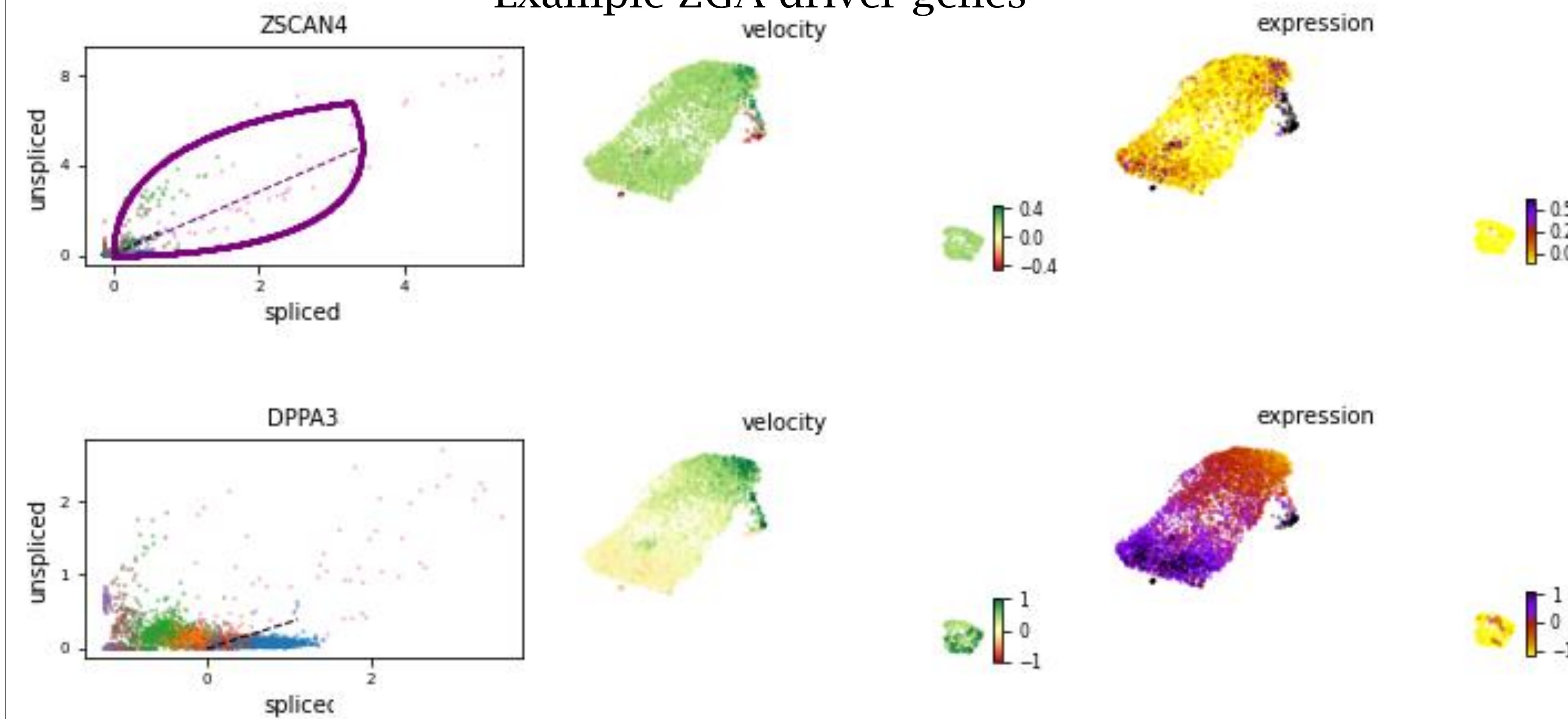


Fig 3: scVelo transcriptional dynamics, velocity (unspliced) and expression (spliced) plots for two driver genes. Includes clustering of each cell

- scVelo infers the gene dynamics (**purple ellipsoid**)
- Velocity derived from unspliced reads indicates **up-** and **down** regulation
- Despite scVelo not inferring a gene dynamics prediction to DPPA3, it's visible that green, pink (8CLCs) and blue clusters show variable unspliced vs spliced reads

Some interesting results

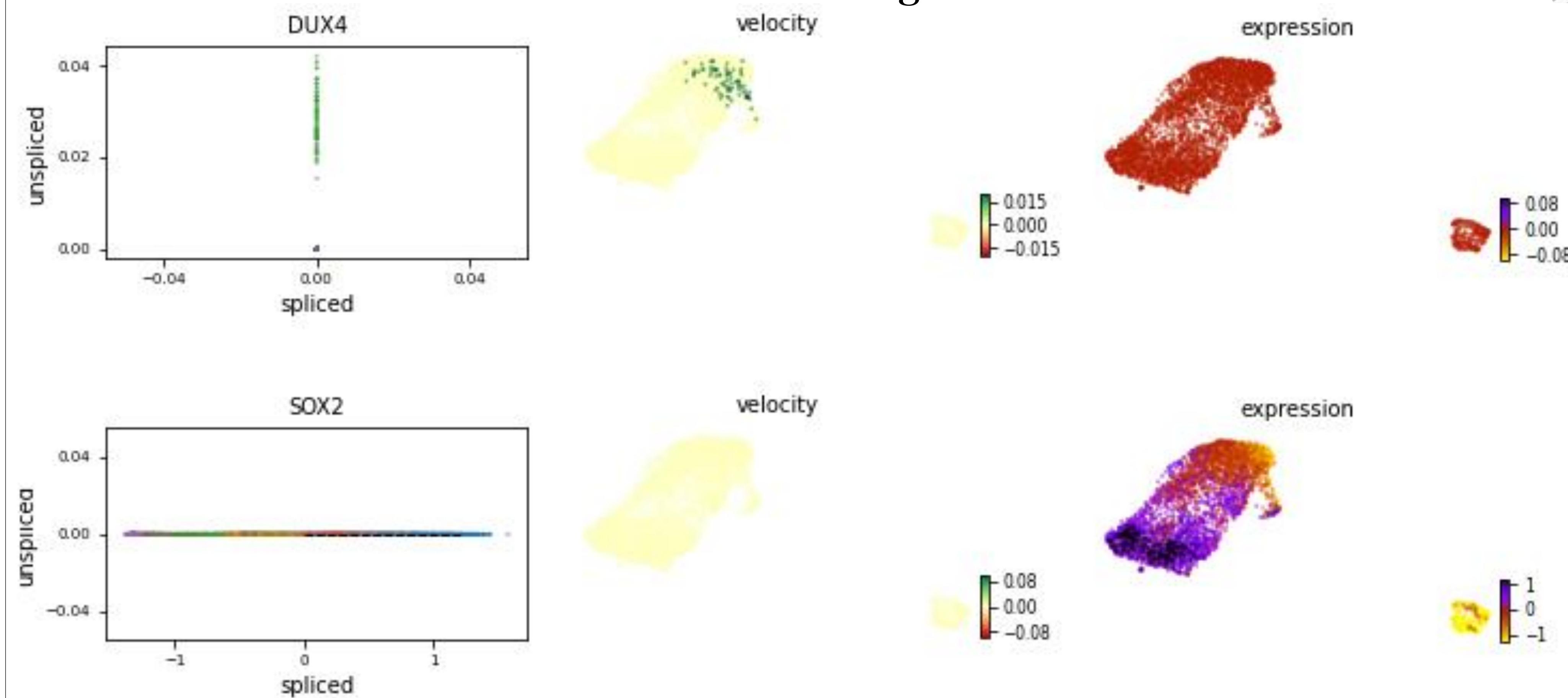


Fig 4: DUX4 lacks velocity (unspliced) reads whilst SOX2 lacks expression (spliced) reads

- DUX4 lacks introns hence interprets all reads as unspliced
- We noticed that the intronic regions for SOX2 may be mislabelled, meaning all reads are interpreted as spliced

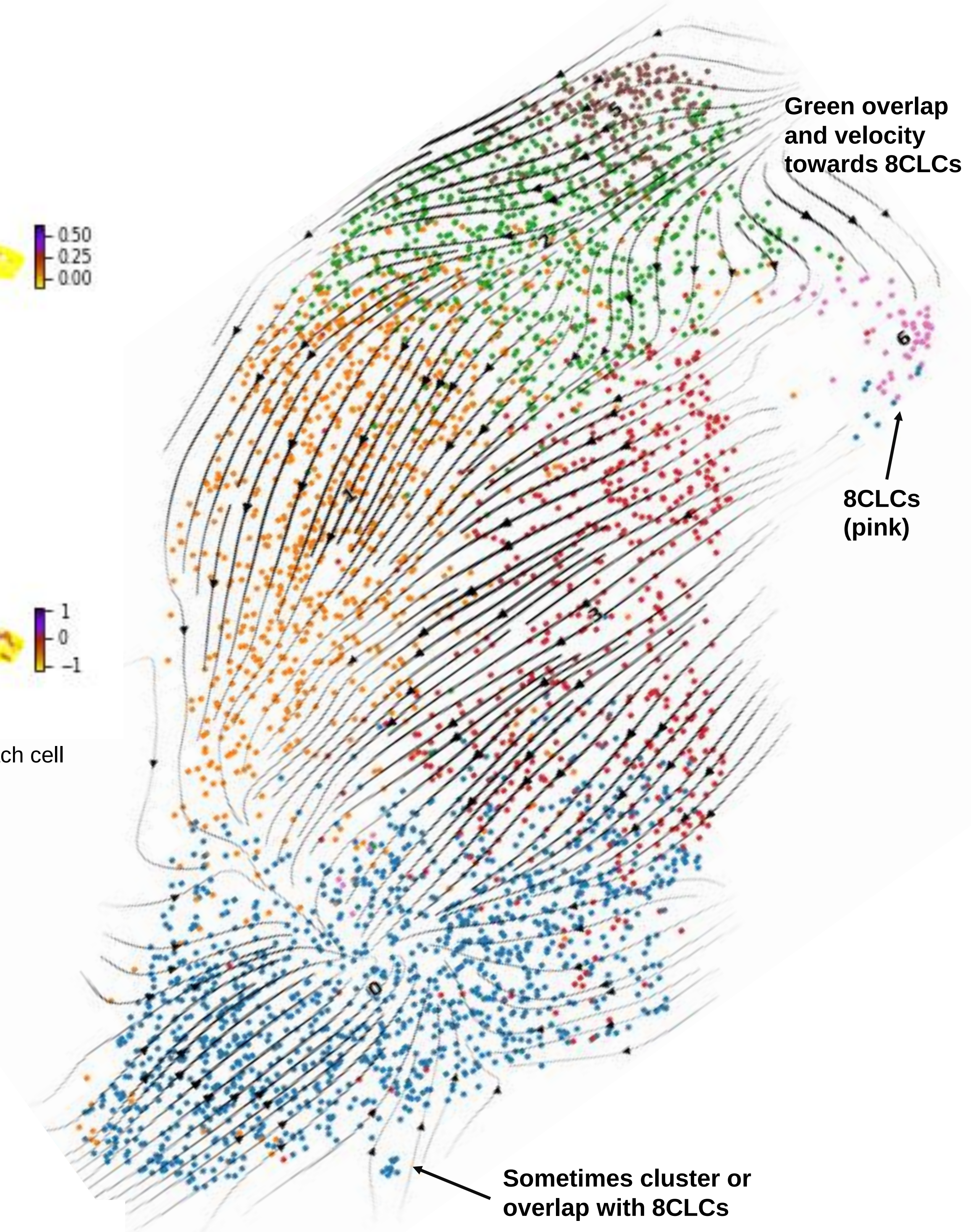


Fig 5: scVelo arrows overlaying UMAP clustering. Pink cluster represents 8CLCs, the rest are naïve ESCs. Green and blue cluster show overlap with pink.

Summary

- 8C-like cells (8CLCs) represent an *in vitro* Zygotic Genome Activation-like (ZGA) state
- RNA velocity allowed us to infer transcriptional dynamics from a **single scRNA sample/timepoint**
- The results indicate that **8CLCs are a transient state** of human naïve ESCs
- They also hint at interesting ZGA splicing dynamics and RNA Velocity limitations